

WINDOWS DIGITAL FORENSICS INVESTIGATION REPORT

NIST CFReDS “Hacking Case” Dataset

Project 1 — Windows Digital Forensics Investigation

Field	Value
Evidence image	SCHARDT.001–SCHARDT.008 (split DD image)
Accompanying acquisition log	SCHARDT.LOG
Operating system	Microsoft Windows XP, x86
Primary user/profile context	Mr. Evil / registered user: Greg Schardt
Primary forensic tool	Autopsy 4.23.1
Supporting tool	Registry Explorer
Archive inspection	WinRAR / working-copy archive inspection
Examination period represented	20–27 August 2004
Report date	24 September 2026
Prepared by	[Investigator name]

Training / Portfolio Report — Evidence-based findings only

Document Control

Item	Value
Case	NIST CFReDS Hacking Case — SCHARDT
Evidence source	Split raw/DD forensic image plus acquisition log
Image handling	Original evidence retained separately; analysis performed in Autopsy
Integrity status	All eight image-part MD5 values verified against SCHARDT.LOG
Primary objective	Identify and document relevant system, execution, network/security-tool, remote-access, browser, and anomaly artifacts
Reporting standard	Evidence strength is explicitly separated from interpretation; unsupported conclusions are not asserted

Contents

1. Executive Summary
2. Scope and Objectives
3. Evidence Description and Integrity
4. Forensic Methodology
5. System and User Context
6. Artifact and Ingest Overview
7. Execution Evidence
8. Browser and Web Activity
9. NetBus Examination
10. VNC / Remote Monitoring Examination
11. Nmap and Network Reconnaissance Artifacts
12. Remote Host Correlation
13. Archive / Interesting Item Examination
14. Extension Mismatch and Encryption Alerts
15. Email, Recent Documents, and Deleted Files
16. Master Timeline
17. Evidence Assessment
18. Limitations
19. Final Conclusion
20. Appendix A — Evidence Hashes
21. Appendix B — Screenshot Index

1. Executive Summary

This report documents a forensic examination of the NIST CFReDS “Hacking Case” Windows XP evidence image represented by eight split DD-image segments, SCHARDT.001 through SCHARDT.008, with the associated SCHARDT.LOG acquisition record. The examination was conducted primarily in Autopsy 4.23.1, with Registry Explorer used for targeted Windows Registry analysis and a working copy used for archive-structure inspection.

The image contains evidence of a Windows XP x86 installation with a user/profile context associated with “Mr. Evil” and a registered user value of Greg Schardt. The recovered artifacts show execution of several networking and security-related utilities, including Cain, Look@LAN, Telnet, WHOIS, NetStumbler, WinPcap components, Ethereal, and mIRC. Browser history contains related downloads, searches, and visits, including activity involving Wardriving.com, NetStumbler, WinPcap, Ethereal, and Look@LAN.

NetBus-related documentation and components were recovered, including NetBus.rtf and NetBus.exe/Patch.exe inside NetBus170.zip. The artifacts establish that NetBus software was stored on the system; however, the examined Prefetch/Run Programs data did not establish execution of NetBus.exe or Patch.exe, and the examined standard Registry and Startup persistence locations did not reveal a corresponding NetBus persistence entry.

VNC software was also recovered, including a standalone vncviewer.exe and an archived copy. The examination did not identify a corresponding vncviewer.exe execution record in the reviewed Prefetch/Run Programs dataset. NmapNT project/configuration data were recovered, including stored scan configurations referencing 192.168.1.* and 192.168.1.1; these configurations do not, by themselves, establish that those scans were executed.

The image also contains cross-artifact references to remote host 4.12.220.254, including a browser history record for file://4.12.220.254/Temp/yng13.bmp and a Recent Documents shortcut referencing the same host and file. The purpose of that access is not established by the recovered artifacts. Automated alerts concerning a possible ZIP bomb and suspected encryption were reviewed but did not, on their own, establish malicious content.

Overall, the examination establishes the presence and, for selected programs, execution of multiple networking/security utilities. It does not establish a complete narrative of why those utilities were used, whether a specific remote-access tool was successfully deployed, or whether any specific unauthorized activity occurred beyond what is directly demonstrated by the artifacts.

2. Scope and Objectives

The examination scope was limited to the recovered disk-image evidence and artifacts derived from it. The objectives were to:

- Validate the integrity of the downloaded split evidence image against the supplied acquisition log.
- Identify the operating system, user/profile context, installed applications, and relevant system artifacts.

- Determine which networking and security-related applications have execution evidence in the recovered artifact set.
- Examine NetBus, VNC, Nmap, and related reconnaissance/security-tool artifacts for presence, execution, and persistence evidence.
- Review browser history, cookies, Recent Documents, deleted-file records, extension mismatches, encryption alerts, interesting items, and email artifacts for corroborating context.
- Develop a master timeline using recovered timestamps without silently correcting or merging timestamp sources.
- Document findings and limitations in a reproducible form suitable for a forensic portfolio.

3. Evidence Description and Integrity

The original evidence consists of eight split image segments. Autopsy was configured with SCHARDT.001 as the image entry point; the remaining sequential segments were automatically associated by Autopsy. SCHARDT.LOG was retained as acquisition documentation and was not treated as an image segment.

The acquisition log identifies the acquisition software as Forensic MD5 V1.77, operating in DD image mode with 650 MB segmenting, LBA address mode, and MD5-file verification. The log identifies an approximately 4.5 GB source drive and a separate 186.3 GB WDC destination/acquisition drive. The physical source-drive characteristics recorded in the log are not treated as the destination drive.

All eight local image-part MD5 values were calculated and matched the values recorded in SCHARDT.LOG. This provides a completed integrity check for the downloaded image segments.

Image segment	MD5 recorded and verified
SCHARDT.001	28A9B613D6EEFE8A0515EF0A675BDEBD
SCHARDT.002	C7227E7EEA82D218663257397679A7C4
SCHARDT.003	EBBA35ACD7B8AA85A5A7C13F3DD733D2
SCHARDT.004	669B6636DCB4783FD5509C4710856C59
SCHARDT.005	C46E5760E3821522EE81E675422025BB
SCHARDT.006	99511901DA2DEA772005B5D0D764E750
SCHARDT.007	99511901DA2DEA772005B5D0D764E750
SCHARDT.008	8194A79A5356DF79883AE2DC7415929F

Evidence handling note: the working project structure was maintained separately from the original evidence. The original evidence directory was treated as read-only and was not used for report exports or modified working copies.

4. Forensic Methodology

The examination used a triage-to-targeted-review workflow:

1. Verified all split evidence-file MD5 values against the acquisition log.
2. Added SCHARDT.001 to a new Autopsy case as a Disk Image or VM File and allowed Autopsy to associate the sequential split segments.
3. Completed ingest processing and reviewed Data Artifacts, Results, Interesting Items, and execution-related results.
4. Reviewed Web History, Web Cookies, Recent Documents, Run Programs/Prefetch, Shell Bags, Metadata, Keyword Hits, Extension Mismatch findings, Encryption Suspected findings, and E-Mail Messages.
5. Extracted Windows Registry hive copies to the working area and examined relevant HKLM/HKCU Run, RunOnce, service, and Startup locations with Registry Explorer.
6. Inspected NetBus and VNC artifacts at the file and archive level; archive files were examined from working copies rather than from the original evidence directory.
7. Reviewed deleted-file results and targeted the list for key names related to NetBus, Patch, VNC, Nmap, Ethereal, Cain, NetStumbler, and Look@LAN.
8. Correlated the strongest artifacts into a master timeline while preserving recovered timestamps exactly as displayed by the source tools.
9. Classified findings as confirmed, corroborating, or inconclusive/not established where appropriate.

Autopsy was the principal forensic platform. Registry Explorer was used only for targeted Registry interpretation. WinRAR was used to inspect nested archive structure in a working copy of `unix_hack.tgz` and to avoid unnecessary extraction of the top-level archive.

5. System and User Context

Attribute	Recovered value
Operating system	Microsoft Windows XP, x86
Windows directory	C:\WINDOWS
Registered user	Greg Schardt
Relevant user/profile context	Mr. Evil
Recovered OS accounts	mr; LOCAL SERVICE; NETWORK SERVICE; SYSTEM

The presence of LOCAL SERVICE, NETWORK SERVICE, and SYSTEM is consistent with standard Windows system/service accounts and is not treated as suspicious solely because those accounts are present.

Browser artifacts contain the profile label “Mr. Evil”. A Yahoo artifact also includes the values `fn=Greg`, `ln=Schardt`, and `id=mrevil2000`. These records provide internal correlation among artifacts but are not treated as independent proof of the real-world identity of a person.

6. Artifact and Ingest Overview

Autopsy produced the following high-level artifact counts during ingest/review:

Artifact category	Recovered count / result
Web History	887
Web Cookies	24
Recent Documents	8
Run Programs	81
Shell Bags	51
Keyword Hits	10,328
Extension Mismatch Detected	9
Interesting Items	6
Encryption Suspected	2
E-Mail Messages	1

Keyword Search was enabled with the Email Addresses list and Solr indexing. No EXIF parser was identified during the review. The report does not infer the presence of additional ingest modules beyond those actually observed in the case results.

7. Execution Evidence

Execution evidence is strongest where a program appears in the recovered Prefetch/Run Programs results. For this report, a Prefetch record is treated as evidence that the executable was launched on the examined Windows installation, while the exact purpose or outcome of the program remains a separate question.

Executable	Recovered timestamp (PKT)	Count	Assessment
CAIN25B45.EXE	20-Aug-2004 20:05:52	1	Execution evidence; beta executable/installer name
WHOIS.EXE	20-Aug-2004 20:10:47	1	Execution evidence
123WASP_SETUP.EXE	20-Aug-2004 20:12:54	1	Execution/launch evidence for installer executable
LOOKATHOST.EXE	25-Aug-2004 20:56:33	1	Execution evidence
MIRC.EXE	25-Aug-2004 21:20:34	3	Execution evidence; multiple recorded launches
TELNET.EXE	26-Aug-2004 20:05:15	1	Execution evidence
LOOKATLAN.EXE	26-Aug-2004 20:06:14	2	Execution evidence
WHOIS.EXE	26-Aug-2004 20:13:57	2	Execution evidence; second recorded launch context

Executable	Recovered timestamp (PKT)	Count	Assessment
NETSTUMBLER.EXE	27-Aug-2004 20:12:35	1	Execution evidence
WINPCAP_3_01_A.EXE	27-Aug-2004 20:15:08	1	Execution evidence for installer
NPF_MGM.EXE	27-Aug-2004 20:15:17	2	Execution evidence; WinPcap-related component
DAEMON_MGM.EXE	27-Aug-2004 20:15:18	3	Execution evidence; WinPcap-related component
ETHEREAL-SETUP-0.10.6.EXE	27-Aug-2004 20:28:36	1	Installer launch evidence
CAIN.EXE	27-Aug-2004 20:33:03	2	Execution evidence
ETHEREAL.EXE	27-Aug-2004 20:34:54	1	Execution evidence

The execution records establish that the listed executables were launched. They do not, by themselves, prove that a specific network operation, password-recovery operation, scan, remote connection, or other action succeeded.

8. Browser and Web Activity

Internet Explorer history provides contextual evidence around several of the executed tools. The history includes searches and visits associated with networking/security topics and software downloads.

Recovered timestamp (PKT)	Browser artifact	Interpretation
20-Aug-2004 15:33:11	MSN search for “hacking”	Browser activity
25-Aug-2004 15:49:42	MajorGeeks Look@LAN download page	Context for Look@LAN acquisition
25-Aug-2004 15:50:14	MajorGeeks direct download URL for Look@LAN installer	Context for installer download
25-Aug-2004 15:51:24	FTP URL for lalsetup250.exe	Context for Look@LAN installer acquisition
25-Aug-2004 16:07:51	Google search for “what is my ip”	Shows IP-address lookup activity; purpose not established
27-Aug-2004 15:09:27	Wardriving.com setup page	Contextual browser activity
27-Aug-2004 15:10:00	NetStumbler downloads page	Context for NetStumbler acquisition
27-Aug-2004 15:11:07	NetStumbler installer URL	Download URL recorded
27-Aug-2004 15:11:33	WinPcap download URL	Download URL recorded
27-Aug-2004 15:11:49	Ethereal installer URL	Download URL recorded
27-Aug-2004 15:24:24	Ethereal installer via FTP mirror	Additional installer acquisition context

The browser timeline and Prefetch timeline contain an apparent approximately five-hour difference for some corresponding events. This report preserves both timestamp streams as recovered and does

not apply a manual offset. The discrepancy may reflect system/browser clock or timestamp semantics, but the underlying cause was not established during this examination.

9. NetBus Examination

NetBus was a primary targeted artifact because the image contains a dedicated NetBus directory and package material. The following evidence was recovered:

- NetBus.rtf at /My Documents/EXPLOITATION/NT/netbus/NetBus.rtf.
- NetBus.exe inside NetBus170.zip.
- Patch.exe inside NetBus170.zip.
- NetBus.bad in a Brutus-related directory containing text referring to NetBUS.
- Additional contextual references to NetBus within network/security tool material and service/signature databases.

NetBus.rtf has an NTFS MFT entry of 10588. Its filesystem record shows a creation time of 20-Aug-2004 20:19:42.021744 PKT, with corresponding MFT/access timestamps in the same 2004 time range. The embedded document metadata contains an older 1998 date. The report treats the filesystem record and embedded metadata as separate timestamp sources and does not describe the document as having been created on this system in 1998.

NetBus-related file	Size	MD5	SHA-256
NetBus.rtf	22,729 bytes	72035134fd7eb8c69377b7bd03004046	de14106b1859f015a0e8212a2f76b21e44869e9558c4288c0aff800e019e2386
NetBus.exe	599,552 bytes	067a8e2d5ccfe6eed1fedfa5d223107	848e12d94d467f2add01403a730e0b4762f594baf70c425fe86b3917c3f0b00b
Patch.exe	494,592 bytes	3542b56a5f9ac8a8c34eb9db6e2f4d00	73a62d6593e3c70b60455299b793bb18d31eddd5f15d04442932c1d3ccb7eb0c

Persistence review: HKLM CurrentVersion Run, RunOnce, and RunOnceEx contained no NetBus-related values. The examined SYSTEM ControlSet001 Services area did not reveal a relevant NetBus/Patch service. HKCU Run for the Mr. Evil profile contained only MSMSGs, and the examined Startup folders contained only desktop.ini. Therefore, no standard NetBus persistence mechanism was identified in the examined locations.

Assessment: NetBus software and documentation were confirmed as stored artifacts. Execution and persistence of NetBus.exe/Patch.exe were not established from the examined Prefetch, Registry, and Startup evidence.

10. VNC / Remote Monitoring Examination

Autopsy identified remote-monitoring-related files including vncviewer.exe artifacts and the normal Windows Remote Desktop client mstsc.exe. The important VNC artifacts were:

- Archived vncviewer.exe within /My Documents/EXPLOITATION/NT/WinVNC/Windows/vnc-3.3.3r7_x86_win32.zip.
- Standalone vncviewer.exe under /My Documents/EXPLOITATION/NT/vnc/vncviewer/vncviewer.exe.
- Standalone VNC viewer file recorded as created, modified, and accessed at approximately 20-Aug-2004 20:19:17 PKT.

Artifact	Size	Hash / identifier
Archived vncviewer.exe	176,128 bytes	MD5 396f4e891335fafb555c298f4534e252
Standalone vncviewer.exe	176,128 bytes	MD5 f7c7f9d6245f5ef0cd419ee140f46fa9; SHA-256 4c6a49fd608ff109efd960eeb07359619bf 9d28fbba9273a7e73e35698a3e249

The complete reviewed Run Programs/Prefetch data did not contain vncviewer.exe. Accordingly, VNC presence is confirmed, while execution of VNC Viewer was not established from the examined execution artifact set.

The SYSTEM registry log identified by Autopsy as a Kaseya VSA-related item was treated as a normal Windows SYSTEM registry log rather than evidence of Kaseya installation or use.

11. Nmap and Network Reconnaissance Artifacts

NmapNT project/configuration material was recovered in both a normal filesystem location and within nmapNTsp1.zip. The stored configuration text included:

- -sS 192.168.1.*
- -sT 192.168.1.1 -d4
- References to source/project components including map.c, osscan.c, targets.c, and wintcpip.c.

These artifacts demonstrate that NmapNT material and scan configuration data were stored on the system. They do not independently establish that the referenced scans were executed. The distinction is important because configuration files and project data may exist without successful execution of the configured operation.

12. Remote Host Correlation

A notable cross-artifact correlation involves the remote host 4.12.220.254. Internet Explorer history contains the entry file:///4.12.220.254/Temp/yng13.bmp dated 26-Aug-2004 15:08:12 PKT. Recent Documents also contains a shortcut to \\4.12.220.254\\TEMP\\yng13.bmp. The same remote host and filename therefore appear in two different artifact classes.

This correlation supports the conclusion that the host/file reference existed in the user environment and browser history. The available evidence does not establish the purpose of the file, the nature of the remote system, or whether the access represented legitimate or unauthorized activity.

13. Archive / Interesting Item Examination

Autopsy flagged `unix_hack.tgz` under `/My Documents/FOOTPRINTING/UNIX/` as a possible ZIP bomb. The file is approximately 73,498,201 bytes and has MD5 `3e1158bec295e21434f5ef4682d33ef4`. The Autopsy message indicated that an archive named `system.tgz` was four levels deep and therefore skipped for processing.

The working copy was examined without extracting the top-level archive. Its structure was:

- `unix_hack.tgz`
- `-> Unix_hack`
- `-> usr`
- `-> hack`
- `-> trinux_dist`
- `-> system.tgz`

The `trinux_dist` area contained multiple Unix/security-related package archives, including `netbase`, `netkit`, `netmap`, `netmon`, `server`, `shell`, `snmp`, and `system` packages. A `tools` area contained items such as `axfr-0.5.2`, `ethereal-0.5.1`, `hunt-1.0`, `hunt-1.3`, `passwd_crackers`, `redirect`, `scanners`, and `trinux_dist` material. The nested `system.tgz` archive was unpacked from the working copy and contained `etc`, `usr`, and `var` directories, including `usr/bin/at` and `crontab`.

Assessment: the automated archive-depth alert is explained by the deep nesting. The artifact demonstrates stored Unix/security toolkit material but does not, by itself, establish malicious behavior or execution of the tools contained within the archives.

14. Extension Mismatch and Encryption Alerts

14.1 Extension Mismatch Findings

Nine extension-mismatch findings were reviewed. Their actual content was examined sufficiently to determine that they did not constitute unexplained executable disguises in this case.

Artifact	Observed content / interpretation
<code>drmv2.sst</code>	OLE/Compound File signature; DRM-related system material
<code>drmv2.lic</code>	OLE/Compound File signature; DRM-related system material
<code>maktoobTracking[1].htm</code>	Cached web graphic/resource; image content
<code>nmapNT.opt</code> (normal location)	NmapNT Visual C++ project/configuration material
<code>nmapNT.opt</code> (inside archive)	NmapNT Visual C++ project/configuration material
<code>Ad468x60.gzf</code>	GIF89a image content despite <code>.gzf</code> extension

Artifact	Observed content / interpretation
SUBACK.BIN	BMP image signature despite .BIN extension
comempty.dat	Installer/database-style OLE content with MSI schema terms
wmimgmt.msc	WMI MMC console configuration content

Assessment: the mismatch alerts were explained by file content or application-specific formats and did not provide an identified disguised executable in the examined set.

14.2 Encryption Suspected Findings

Two oembios.bin files, under WINDOWS/system32 and dllcache, were flagged by Autopsy for suspected encryption because of extremely high entropy (reported as 7.999987). High entropy alone is not sufficient to establish encryption, obfuscation, or malicious content. The files were therefore retained as an automated unresolved alert rather than elevated to a substantive finding.

15. Email, Recent Documents, and Deleted Files

15.1 Email Message

Autopsy reported one email message. The recovered content is technical correspondence concerning porting the fping utility to VMS, including source-code modification, compilation/linking, privileges, and usage examples. No sender, recipient, or date headers were available in the reviewed artifact presentation sufficient to attribute the message to a particular person or to establish execution of fping.

Assessment: contextual technical correspondence; not execution evidence.

15.2 Recent Documents

Shortcut	Target / stored reference
GhostWare.lnk	D:\Drivers\GhostWare
Anonymizer.lnk	D:\Drivers\Anonymizer
keys.lnk	D:\Drivers\Anonymizer\keys.txt
Receipt.lnk	D:\Drivers\GhostWare\Receipt.rtf
Temp on m1200 (4.12.220.254).lnk	No preferred path recorded
yng13.lnk	\\4.12.220.254\TEMP\yng13.bmp
channels (2).lnk	C:\Program Files\mIRC\channels
channels.lnk	C:\Program Files\mIRC\channels\channels.txt

The Recent Documents data is best treated as a correlation source rather than proof that each referenced file was opened manually during a specific session.

15.3 Deleted Files

The deleted-file results were reviewed in full. The list contained numerous zero-byte unallocated records and Windows system/cache remnants. Notable non-security files included ncreadme.txt, NDELFILE.EXE, NOPNFILE.EXE, and XPOST.DAT. A targeted name review did not identify deleted copies of netbus, patch, vnc, nmap, ethereal, cain, netstumbler, or lookatlan that would materially change the primary findings.

Assessment: deleted-file artifacts did not produce additional direct evidence concerning execution of the primary NetBus or VNC artifacts.

16. Master Timeline

The following timeline consolidates the strongest recovered events. Timestamps are preserved exactly as presented by the respective artifact source. Browser history and Prefetch timestamps are intentionally not normalized into a common clock.

Recovered time (PKT)	Source	Event
20-Aug 15:33:11	IE History	MSN search for “hacking”
20-Aug 20:05:52	Prefetch	CAIN25B45.EXE — count 1
20-Aug 20:10:47	Prefetch	WHOIS.EXE — count 1
20-Aug 20:12:54	Prefetch	123WASP_SETUP.EXE — count 1
20-Aug 20:19:42	NTFS	NetBus.rtf filesystem creation record
25-Aug 15:49:42	IE History	MajorGeeks Look@LAN download page
25-Aug 15:50:14	IE History	Look@LAN installer download URL
25-Aug 15:51:24	IE History	FTP lalsetup250.exe URL
25-Aug 16:07:51	IE History	Google search for “what is my ip”
25-Aug 20:56:33	Prefetch	LOOKATHOST.EXE — count 1
25-Aug 21:20:34	Prefetch	MIRC.EXE — count 3
26-Aug 15:08:12	IE History	file:///4.12.220.254/Temp/yng13.bmp
26-Aug 20:05:15	Prefetch	TELNET.EXE — count 1
26-Aug 20:06:14	Prefetch	LOOKATLAN.EXE — count 2
26-Aug 20:13:57	Prefetch	WHOIS.EXE — count 2
27-Aug 15:09:27	IE History	Wardriving.com setup page
27-Aug 15:10:00	IE History	NetStumbler downloads page
27-Aug 15:11:07	IE History	NetStumbler installer URL
27-Aug 15:11:33	IE History	WinPcap download URL
27-Aug 15:11:49	IE History	Ethereal installer URL
27-Aug 20:12:35	Prefetch	NETSTUMBLER.EXE — count 1
27-Aug 20:15:08	Prefetch	WINPCAP_3_01_A.EXE — count 1

Recovered time (PKT)	Source	Event
27-Aug 20:15:17	Prefetch	NPF_MGM.EXE — count 2
27-Aug 20:15:18	Prefetch	DAEMON_MGM.EXE — count 3
27-Aug 20:28:36	Prefetch	ETHEREAL-SETUP-0.10.6.EXE — count 1
27-Aug 20:33:03	Prefetch	CAIN.EXE — count 2
27-Aug 20:34:54	Prefetch	ETHEREAL.EXE — count 1

17. Evidence Assessment

Findings are assessed using the following terms: Confirmed = directly established by the examined artifact; Corroborating = supports another finding but does not independently establish the activity; Inconclusive / not established = the artifact exists but the available evidence does not establish the proposed activity.

Finding	Assessment	Basis
Execution of Cain, Look@LAN, Telnet, WHOIS, NetStumbler, WinPcap components, Ethereal, and mIRC	Confirmed	Prefetch/Run Programs records provide execution evidence.
Browser downloads/visits associated with Look@LAN, NetStumbler, WinPcap, and Ethereal	Corroborating	History supports acquisition/context but does not describe every subsequent action.
NetBus software/documentation stored on the image	Confirmed	NetBus.rtf, NetBus.exe, and Patch.exe artifacts recovered.
NetBus.exe / Patch.exe execution	Not established	No corresponding execution record identified in reviewed Prefetch/Run Programs artifacts.
Standard NetBus persistence	Not established	No relevant Run/RunOnce/service/Startup mechanism identified in examined locations.
VNC software stored on the image	Confirmed	Archived and standalone vncviewer.exe artifacts recovered.
VNC Viewer execution	Not established	No vncviewer.exe Prefetch/Run Programs record identified.
NmapNT configuration/project material	Confirmed	Files include stored scan configurations and source/project material.
Execution of referenced Nmap scans	Inconclusive	Configuration data alone does not establish execution.
Remote host 4.12.220.254 referenced in multiple artifacts	Corroborating	IE history and Recent Documents reference the same host/file.
Possible ZIP bomb alert	Resolved as non-dispositive	Deep archive nesting explains the processing alert; maliciousness not established.
Encryption Suspected alert on oembios.bin	Inconclusive	High entropy alone does not establish encryption or malicious content.
Extension mismatch alerts	Resolved	Content examination provided benign/format-specific explanations.

Finding	Assessment	Basis
Deleted copies of primary target tools	Not recovered	Targeted deleted-file review did not identify deleted NetBus/Patch/VNC/Nmap/Ethereal/Cain/NetStumbler/Look@LAN copies.

18. Limitations

- The examination is limited to the supplied disk image and artifacts derived from it. No live-system acquisition, RAM image, external network captures, or independent endpoint telemetry were available.
- The browser and Prefetch timestamp streams contain an apparent offset for some related events. The cause was not established, so the report preserves recovered timestamps rather than applying an unverified correction.
- Prefetch establishes launch evidence but does not by itself prove successful completion of an operation or the exact intent behind the launch.
- Installed software, archives, configuration files, and documentation establish presence/storage, not necessarily use.
- The Registry and Startup persistence review covered the standard locations examined during this investigation. Absence from those locations does not exclude every possible persistence mechanism.
- The internal profile name “Mr. Evil” and the Yahoo values linking Greg, Schardt, and mreuil2000 provide artifact correlation but are not treated as independent identity verification.
- Remote-host references do not, by themselves, establish ownership, location, legitimacy, or purpose of the remote system.
- The analysis did not attempt malware reverse engineering or execution of recovered binaries. Such actions were outside the scope of the disk-image triage performed here.

19. Final Conclusion

The forensic examination of the NIST CFReDS “Hacking Case” image establishes a Windows XP x86 environment containing and executing multiple networking and security-related utilities. The strongest execution evidence comes from Prefetch/Run Programs artifacts for Cain, Look@LAN, Telnet, WHOIS, NetStumbler, WinPcap-related components, Ethereal, and mIRC. Internet Explorer history provides additional contextual evidence of visits, searches, and software acquisition involving several of these tools.

The examination confirmed the presence of NetBus documentation and executable components, as well as VNC software. However, execution of NetBus.exe/Patch.exe and execution of vncviewer.exe were not established from the reviewed execution artifacts. The standard persistence mechanisms examined did not reveal a NetBus startup or service entry. NmapNT project/configuration material

was recovered, but the stored scan configurations do not independently establish execution of the referenced scans.

A cross-artifact reference to remote host 4.12.220.254 was identified in both browser history and Recent Documents. This is a useful correlation, but the purpose of the access is not established by the available evidence. Automated archive and entropy alerts were investigated and did not, by themselves, establish maliciousness or encryption.

Accordingly, the evidence supports a finding that the examined system contained and, for selected applications, executed a range of networking/security software. The evidence does not support a broader unsupported conclusion about a specific actor, intent, successful unauthorized access, or complete attack sequence. Any such determination would require additional corroboration outside the examined artifacts.

20. Appendix A — Evidence Hashes

A.1 Split Image MD5 Verification

File	MD5
SCHARDT.001	28A9B613D6EEFE8A0515EF0A675BDEBD
SCHARDT.002	C7227E7EEA82D218663257397679A7C4
SCHARDT.003	EBBA35ACD7B8AA85A5A7C13F3DD733D2
SCHARDT.004	669B6636DCB4783FD5509C4710856C59
SCHARDT.005	C46E5760E3821522EE81E675422025BB
SCHARDT.006	99511901DA2DEA772005B5D0D764E750
SCHARDT.007	99511901DA2DEA772005B5D0D764E750
SCHARDT.008	8194A79A5356DF79883AE2DC7415929F

A.2 Key Recovered File Hashes

Artifact	MD5	SHA-256
NetBus.rtf	72035134fd7eb8c69377b7bd03004046	de14106b1859f015a0e8212a2f76b21e44869e9558c4288c0aff800e019e2386
NetBus.exe	067a8e2d5ccfe6eed1fedfa5d223107	848e12d94d467f2add01403a730e0b4762f594baf70c425fe86b3917c3f0b00b
Patch.exe	3542b56a5f9ac8a8c34eb9db6e2f4d00	73a62d6593e3c70b60455299b793bb18d31edd5f15d04442932c1d3ccb7eb0c
Standalone vncviewer.exe	f7c7f9d6245f5ef0cd419ee140f46fa9	4c6a49fd608ff109efd960eeb07359619bf9d28fbaa9273a7e73e35698a3e249
Archived vncviewer.exe	396f4e891335fafb555c298f4534e252	Not recorded in the reviewed notes
unix_hack.tgz	3e1158bec295e21434f5ef4682d33ef4	Not recorded in the reviewed notes

21. Appendix B — Screenshot Index

The final portfolio package should place selected screenshots under 04_Screenshots. The following index can be used to name or caption the images consistently:

ID	Screenshot	Purpose / caption
S01	Autopsy case tree / evidence source	Shows SCHARDT.001 loaded with split-image association.
S02	Ingest results overview	Shows key artifact counts after ingest.
S03	Run Programs / Prefetch — security tools	Cain, Look@LAN, Telnet, WHOIS, NetStumbler, WinPcap, Ethereal.
S04	IE Web History — tool acquisition	Look@LAN, NetStumbler, WinPcap, Ethereal, Wardriving activity.
S05	NetBus.rtf details	Path, size, timestamps, and content summary.
S06	NetBus archive members	NetBus.exe and Patch.exe inside NetBus170.zip.
S07	Registry — HKLM Run / RunOnce / RunOnceEx	Shows examined persistence locations with no NetBus value.
S08	Registry — SYSTEM services	Shows reviewed service area with no relevant NetBus/Patch service.
S09	VNC artifacts	Standalone and archived vncviewer.exe evidence.
S10	NmapNT configuration	Shows stored 192.168.1.* / 192.168.1.1 scan configuration.
S11	Remote host correlation	Browser history and Recent Documents reference 4.12.220.254 / yng13.bmp.
S12	unix_hack.tgz structure	Shows nested archive structure and security-tool packages.
S13	Extension mismatch review	Representative file signatures/content demonstrating benign format mismatches.
S14	Encryption Suspected	Shows oembios.bin alert and entropy value.
S15	Deleted-file review	Shows targeted deleted-file findings / absence of primary target copies.

End of report